DSP

Forever Notes

HW

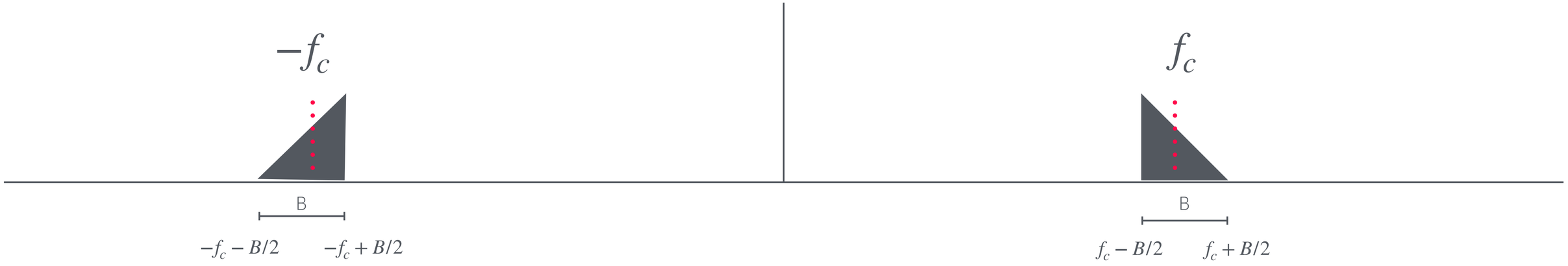
Chapter 2

Sampling

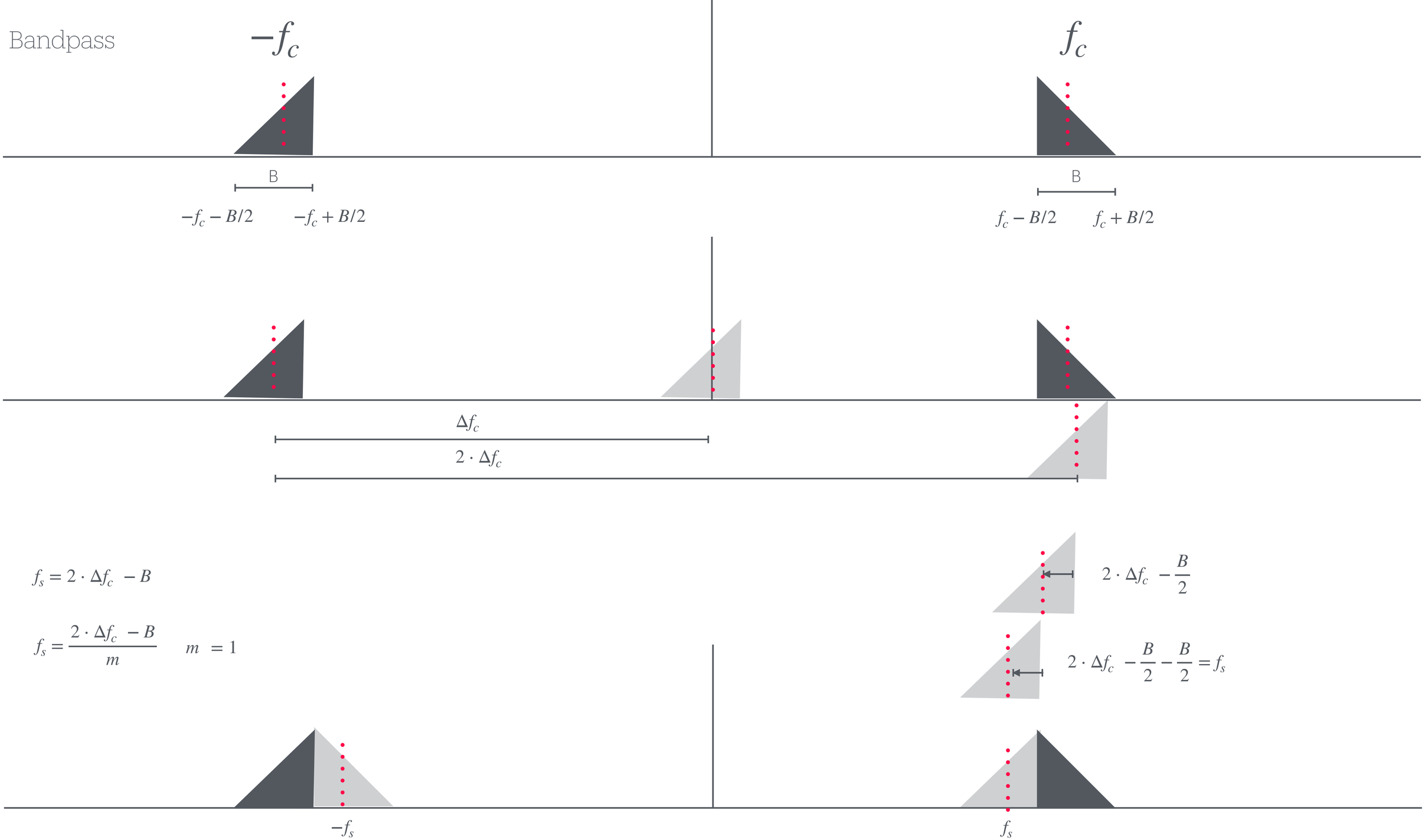
Sampling Architecture

$$Sampling\ Rule : f_s \geq 2B$$

Bandpass



Bandpass



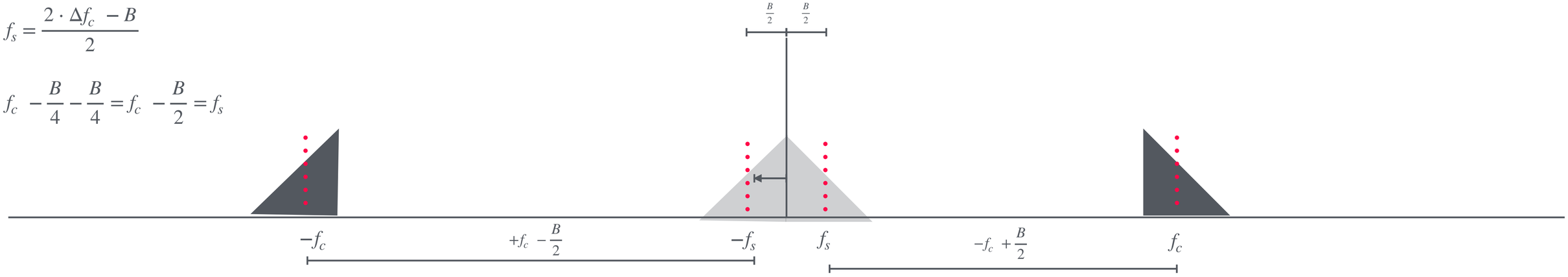
$$f_s = 2 \cdot \Delta f_c - B$$

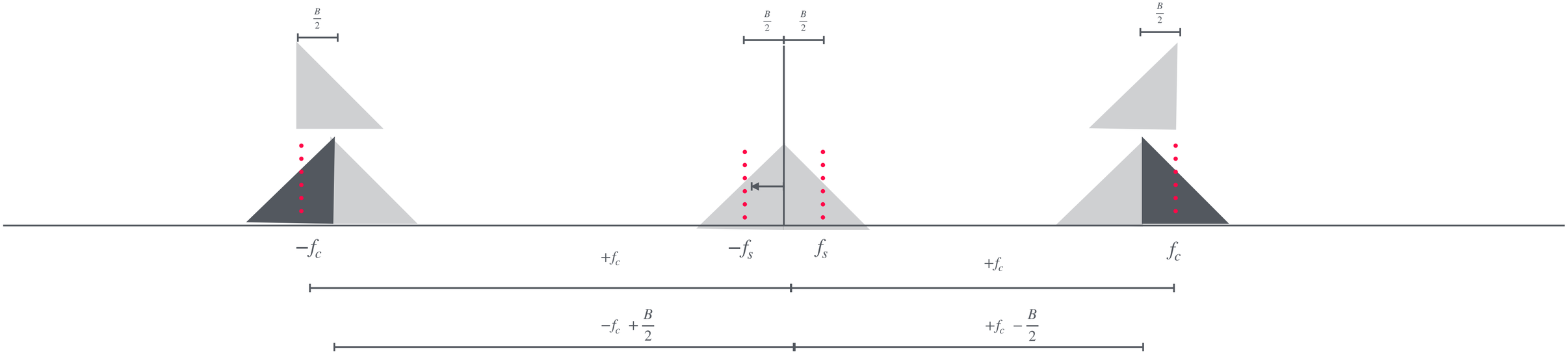
$$f_s = \frac{2 \cdot \Delta f_c - B}{m} \qquad m = 1$$

Bandpass

$$f_s = \frac{2 \cdot \Delta f_c - B}{2}$$

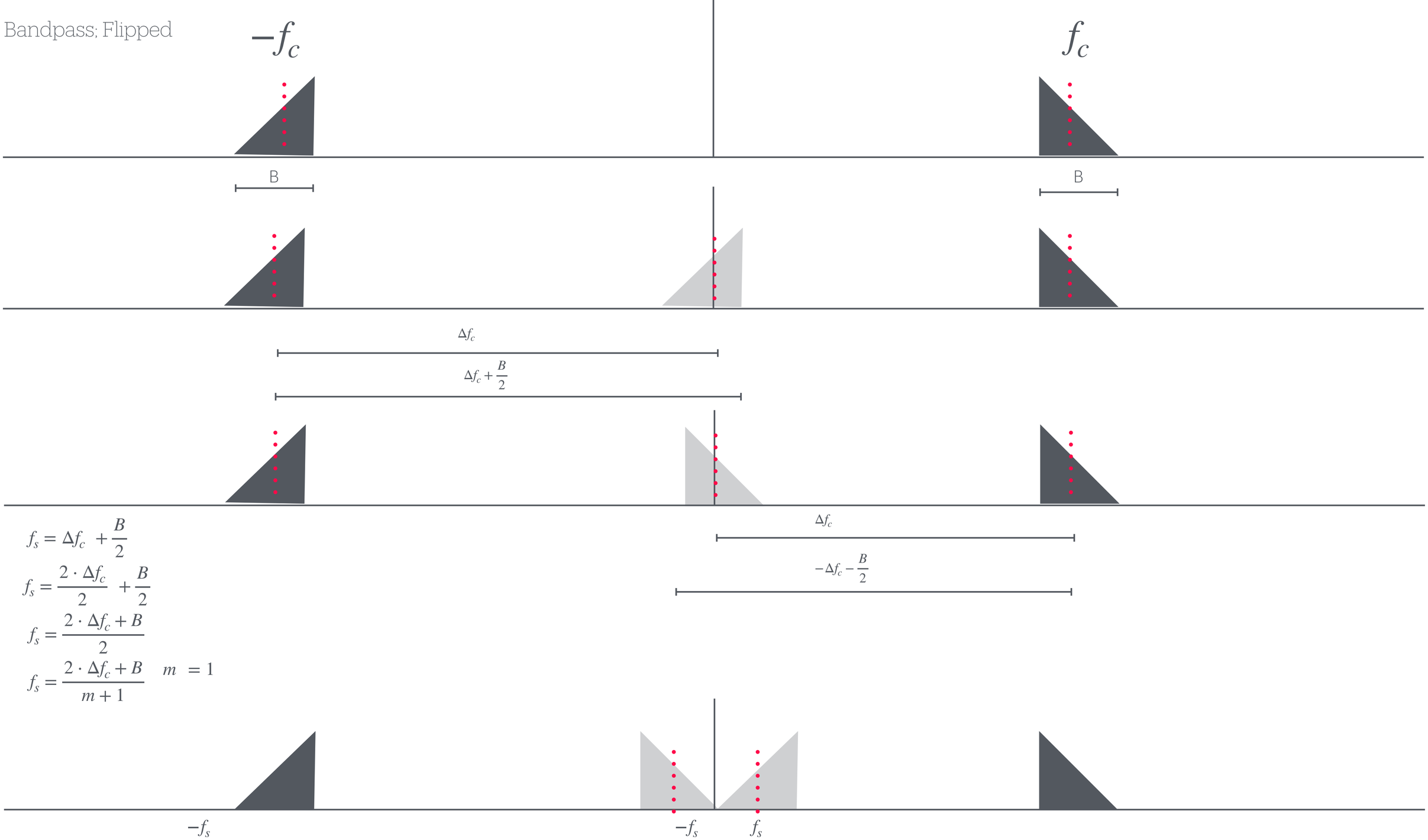
$$f_c - \frac{B}{4} - \frac{B}{4} = f_c - \frac{B}{2} = f_s$$





$$f_s = \frac{2 \cdot \Delta f_c - B}{m} \qquad m = 2$$

Bandpass; Flipped



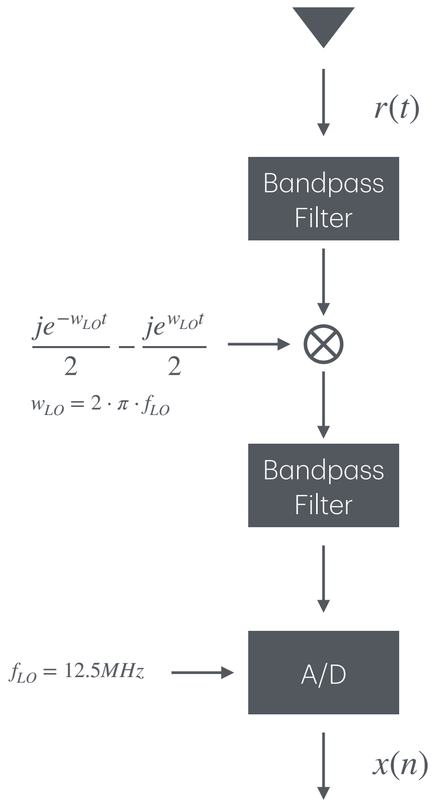
$$f_s = \Delta f_c + \frac{B}{2}$$

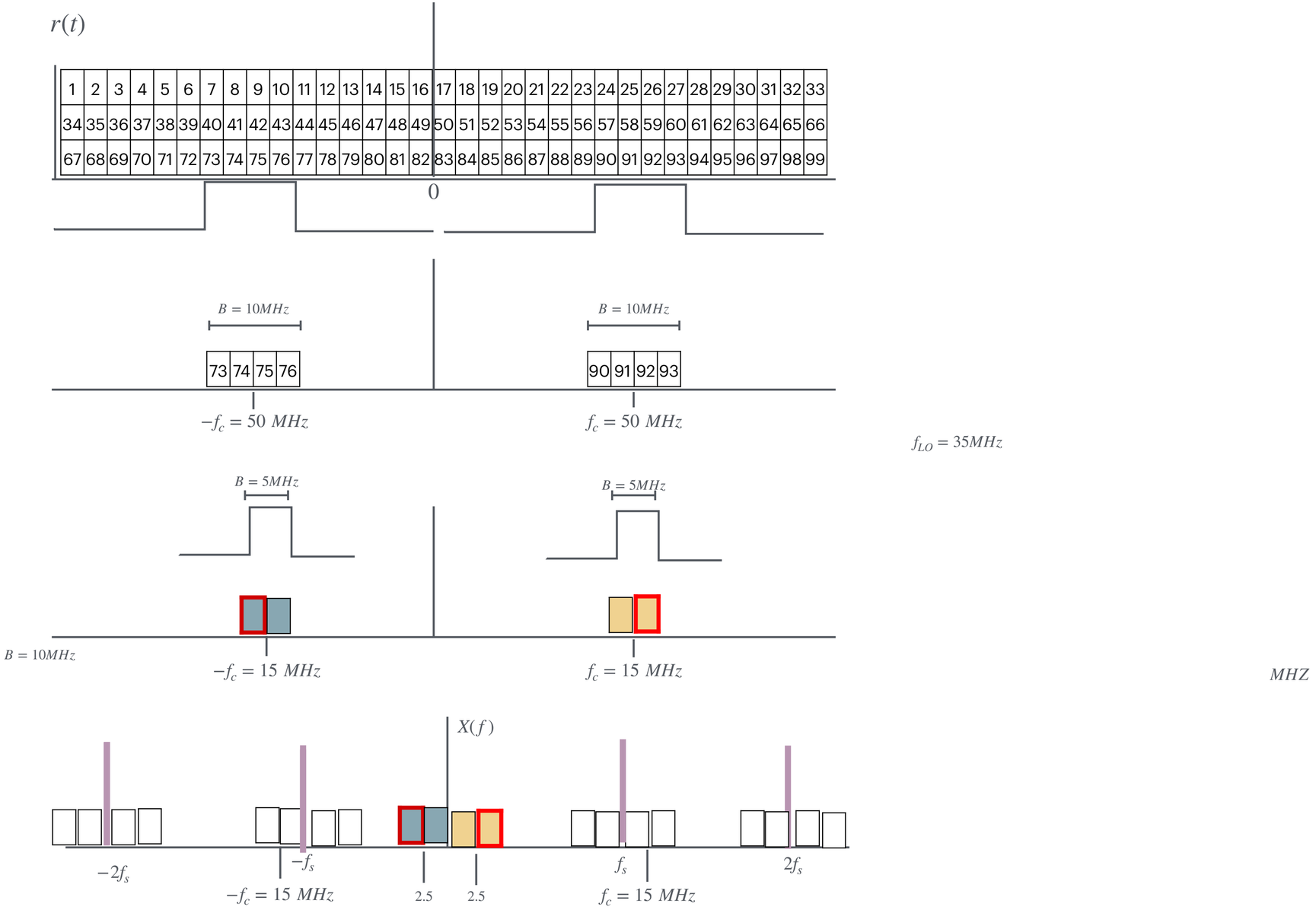
$$f_s = \frac{2 \cdot \Delta f_c}{2} + \frac{B}{2}$$

$$f_s = \frac{2 \cdot \Delta f_c + B}{2}$$

$$f_s = \frac{2 \cdot \Delta f_c + B}{m+1} \quad m = 1$$

Example


$r(t)$
Bandpass Filter
$\frac{je^{-w_{LO}t}}{2} - \frac{je^{w_{LO}t}}{2}$
$w_{LO} = 2 \cdot \pi \cdot f_{LO}$
Bandpass Filter
$f_{LO} = 12.5MHz$
A/D
$x(n)$


$r(t)$
0
$B = 10MHz$
$B = 10MHz$
$-f_c = 50\ MHz$
$f_c = 50\ MHz$
$f_{LO} = 35MHz$
$B = 5MHz$
$B = 5MHz$
$B = 10MHz$
$-f_c = 15\ MHz$
$f_c = 15\ MHz$
MHZ
$X(f)$
$-2f_s$
$-f_s$
$-f_c = 15\ MHz$
2.5
2.5
f_s
$f_c = 15\ MHz$
$2f_s$